뇌-마음-행동 개별연구 승인 신청서

신청자	성명	윤정원
	학과(전공)	의학과
	학번	2022-18923
	연락처	이메일: juliaaayun@snu.ac.kr 핸드폰: 010-7722-6287
개별연구 주제	Understanding EEG signals through Foundation Models; probing critical features with traditional methods and seeking generalizability by representational similarity analysis between models, subjects, and tasks	
개별연구 개요	EEG signal analysis has conventionally relied on the heuristics and expertise of clinicians, gained through years of training. Recent deep learning techniques, however, offer the possibility of uncovering deeper insights into these signals through latent space representations. Despite this potential, current research on decoding EEG signals is limited to subject and task specific learning, inevitably reproducing subtly different models with slightly different recording conditions, confining the range of general understanding on brain signals. To overcome such limitations, we propose the need of foundation models on EEG signals just like large language models in natural language processing. This study leverages the Codabench EEG Challenge as a platform to build and evaluate such models utilizing diverse structures for neural signal decoding. Beyond performance, this study will investigate which aspects of traditional methods remain relevant or not by modifying the models' architecture. For instance, following questions are possible; Will frequencies of typical bands still be key features in foundation models?; How significant is local spatial information such as channel positions, for denoising and decoding across diverse tasks?; Does the number of critical tokens for attention rely on autocorrelation lengths which vary by subject status? Lastly, by analyzing embeddings learned by high performing models, this study seeks to identify representational similarity between model types, subjects, and tasks. Such comparison between models will allow understanding on which component, optimization or representation, contributes to performance gaps. With subject-task analysis, we could probe the generalizability of foundation models. Through this approach, we aim to enhance decoding performance of EEG signals, allowing subject-task generalization, and further expand our scientific understanding of their neural dynamics.	

위와 같이 뇌-마음-행동 개별연구를 신청합니다.

2025년 9월 18일

신청자: 윤정원 (서명)

지도교수	성명	차지욱
	소속학과	심리학과

위 학생의 개별연구를 승인합니다.

2025년 9월 19일

지도교수: 차지욱